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Design and construction of a low-cost 3D-printed portable LIBS system

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Abstract. Laser-Induced Breakdown Spectroscopy (LIBS) is a widely used non-destructive analysis technique, which is however costly. In this paper we present the development of a low-cost 3D-printed portable LIBS system designed and built with a Technology Readiness Level 5 (TRL 5), i.e., technology validated in a relevant environment. The proposed LIBS system is compact and includes a passive Q-switch Nd^{3+}:YAG pulsed laser, a handheld measuring device, a power supply, an optical spectrometer and a processing software. The electronic control system guarantees a stable laser triggering, thanks to the use of optical connections (optocouplers) avoiding wired electrical connections of the circuits. In addition, we found the correct energy characteristics of the passive Q-switch regime, to generate a single laser pulse, thus making the system operate more efficiently. Our low-cost 3D-printed portable LIBS system was validated with several tests with real samples of atomic elements (i.e., lithium, copper, silver, gold and lead).

1. Introduction

Analysis methods based on Atomic Emission Spectroscopy have been used for a long time and have evolved over the years towards different applications, while keeping the same basic concept: Atoms emit characteristic spectral lines when excited by some energy source; if such spectral lines can be observed in any sample, this confirms the presence of the atom even though the sample contains other elements. Also, the measured intensity of the spectral lines provides quantitative information of the element in the sample [1]. Atomic Emission Spectroscopy uses an energy source to convert a portion of matter into plasma and forces its atoms to emit photons. Spectrometers analyze the emitted radiation in order to determine the atomic composition of the sample.

The main component of the Atomic Emission Spectroscopy is the energy source used for sample excitation. Some of the most commonly used atomic emission excitation sources are the *spark electrode* and the *inductively coupled plasma* (ICP), which reach temperatures of 50000 K and 10000 K, respectively [2].

With the development of laser technology in the early 60s, several researchers investigated its use in atomic emission spectroscopy [3], and rapidly lasers became an important excitation source for atomic emission analysis [4]. In 1963, Laser-Induced Breakdown Spectroscopy - LIBS [5] introduced the analytical use of plasma generated by laser radiation on surfaces. Currently, several types of pulsed

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lasers are used in LIBS systems, with the Nd^{3+}:YAG type being the most common. The temperature of the laser-generated plasma varies between 8000 K and 12000 K [2].
LIBS technique has shown several advantages compared to other methods (e.g., Atomic Absorption Spectroscopy - AAS or Inductively Coupled Plasma Mass Spectrometry - ICP-MS), such as its non-destructive analysis capability, in-situ measurement capability and portability. The non-destructive analysis capability allows the LIBS technique to be used in hostile and hazardous environments for humans, and its in-situ measurement capability and portability allows it to be used in the field, thus reducing analysis time [6].
The applications of LIBS covers diverse areas, including agriculture and food [7], explosives detection [8], analysis of terrestrial and extraterrestrial minerals and rocks [9], nuclear fusion technologies [10], industrial applications [11], archaeology [12], human and animal health [13], among many others.
Although LIBS has important advantages over other atomic spectral analysis techniques, it is extremely expensive. In this sense, our long-term research effort focuses on the development of low-cost spectrometers and pulsed lasers, integrated into a functional portable LIBS system.
In this paper, we present the development of a low-cost 3D-printed portable LIBS system and a control software for in-situ and real-time detection of atomic samples, i.e., an equipment with a Technology Readiness Level 5 (TRL 5) [14]. This work is based on previous research results in the development of a Nd^{3+}:YAG pulsed lasers [15] and low-cost optical spectrometers [16,17,18] which had TRL 4 category (i.e., technology validated in a laboratory). The contribution of this paper, compared to previous work, can be summarized as follows:

- The optimization of the control electronic system to avoid pulse losses during laser triggering.
- Energy control of the optical pumping to ensure a single pulse in passive Q-switch regime.
- The design and construction of a robust 3D-printed and portable LIBS system.

The rest of the paper is structured as follows: Section 2 describes the mechanical design and 3D printing. Section 3 presents the design and development of the electronic power supply of the laser and the processing software. In Section 4, we describe the results obtained with real samples and Section 5 concludes the paper.

2. Mechanical design and 3D printing

Our proposed LIBS system consists of (a) a high-power pulsed laser that generates the plasma in the sample, (b) an optical spectrometer, and (c) a light collection system that collects the light emitted by the plasma and sends it through an optical fiber to the spectrometer. The spectrometer is responsible for the analysis of the different wavelengths produced by the plasma light.
In order to allow our device to operate in real-world field conditions, we need to ensure a robust and resistant mechanical structure, so as to keep the internal components fixed. In addition, those internal components, i.e, the electronics, optical (pulsed laser and signal collection system) and spectrometric (optical spectrometer), must be designed to be as compact as possible. Our approach is to use robust and low-cost 3D printing material, which eases the design and manufacturing of the portable LIBS prototype.

2.1. Design of the handheld measuring device

The LIBS handheld measuring device has two main functions: (a) it contains the pulsed Nd^{3+}:YAG laser and focalizes the laser radiation into the sample, and (b) it contains an optical system for collecting the light generated by the plasma that is carried to the spectrometer through an optical fiber.
This mechanical part must be as ergonomic as possible in order to allow the user to simply approach and point the sample to be analyzed, without requiring any special alignment, and trigger the laser by pushing a button. This solution allows efficient collection of the light of the plasma in non-homogeneous surfaces.

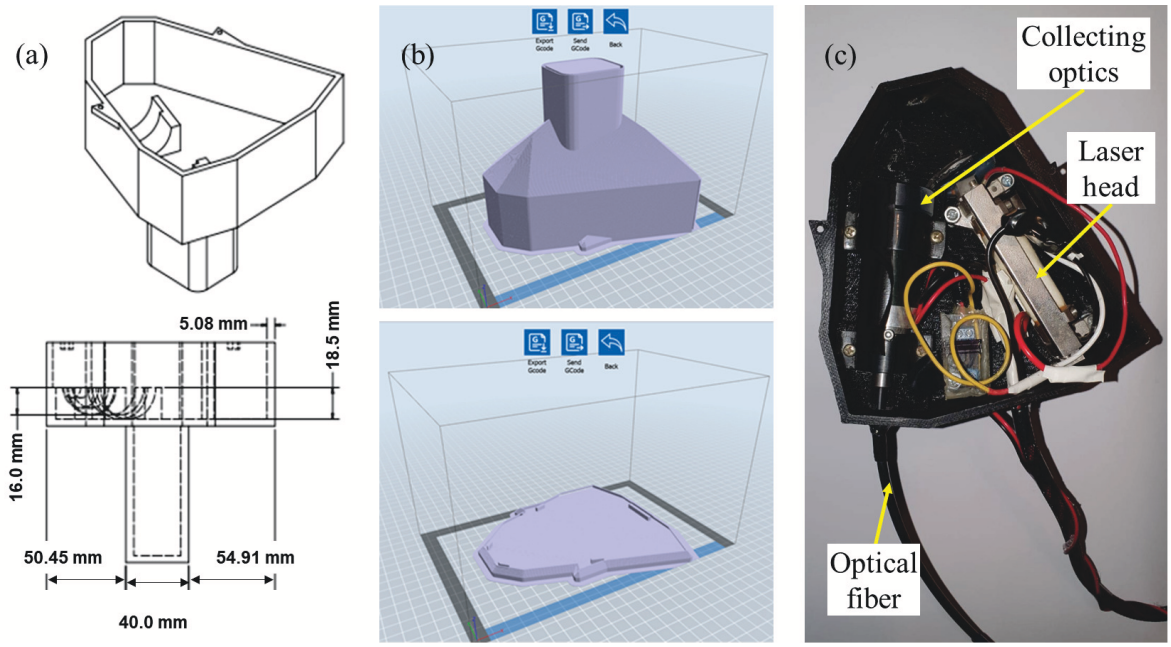

Figure 1. Mechanical design of the handheld structure (a), SolidWorks design (b) and the mounted handheld measuring device (c).

Figure 1(a) shows the design of the robust mechanical structure of the handheld measuring device, Figure 1(b) shows the 3D design in SolidWorks ready to be manufactured, and Figure 1(c) shows the assembled handheld measuring device. The mechanical structure was 3D-printed using black PLA filament to avoid, as much as possible, laser beam reflections for eye safety reasons.

The excitation light source is a small Nd^{3+}:YAG laser head which emits laser radiation at 1064 nm. It includes a flash lamp with external trigger, a YAG rod, a passive Q-switch crystal, and two mirrors, all assembled in a 25x25x105 mm compact metal case. The energy of each pulse in the passive Q-switch regime is of 20 mJ with a duration of 40 ns (see Figure 2(a)), with an energy input of 16 J at 100 µs flash lamp pulse duration. The measurement of the pulse duration was made with a high speed photodetector (11HSP-FS1 with a wavelength range of 400-1100 nm) and a 200 MHz bandwidth Tektronix oscilloscope. For the energy monopulse measurement, we used a portable laser power and energy meter (StarLite of Ophir). The maximum trigger frequency of the monopulse emission in the Q-switch regime was 0.14 Hz. This frequency was used because the laser system is only cooled by convection with the ambient air.

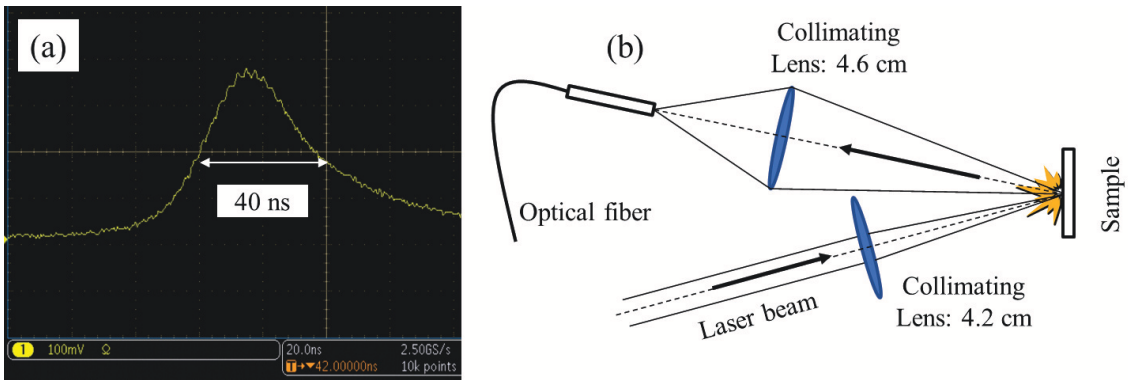

Figure 2. Oscillogram of the temporal duration of the laser pulse in the passive Q-switch regime (a) and the optical light collecting scheme (b).

To generate the plasma on the sample, the laser beam is focalized with a collimating lens (focus length: 4.2 cm). Figure 2(b) shows the optical scheme for collecting the light generated by the plasma, which is composed by a second collimating lens (focus length: 4.6 cm) and an optical fiber. The collimating lens is responsible for collecting all the light from the plasma, which is transmitted to the spectrometer through the optical fiber.

2.2. Design of the mechanical structure of the optical spectrometer

Based on previous work [19], we designed an optical spectrometer with a reflection configuration. It uses two compact optical objective lenses, a diffraction grating of 1200 lines/mm, and a digital monochromatic camera (Chameleon CCD digital camera) as opto-electronic sensor. The spectral resolution (measured with the method described in [16]) of the developed optical spectrometer is 0.27 nm and a spectral window of 76.9 nm.

Figure 3(a) shows the design of the mechanical structure of the optical spectrometer, Figure 3(b) shows the designs in SolidWorks, and Figure 3(c) shows the assembled optical spectrometer. A variable slit was additionally mounted on this structure to control the light coming from the optical fiber that carries the signal produced by the plasma generated in the sample. Similarly, a 3D-printed structure was built to hold the laser power supply and the electronic control circuit.

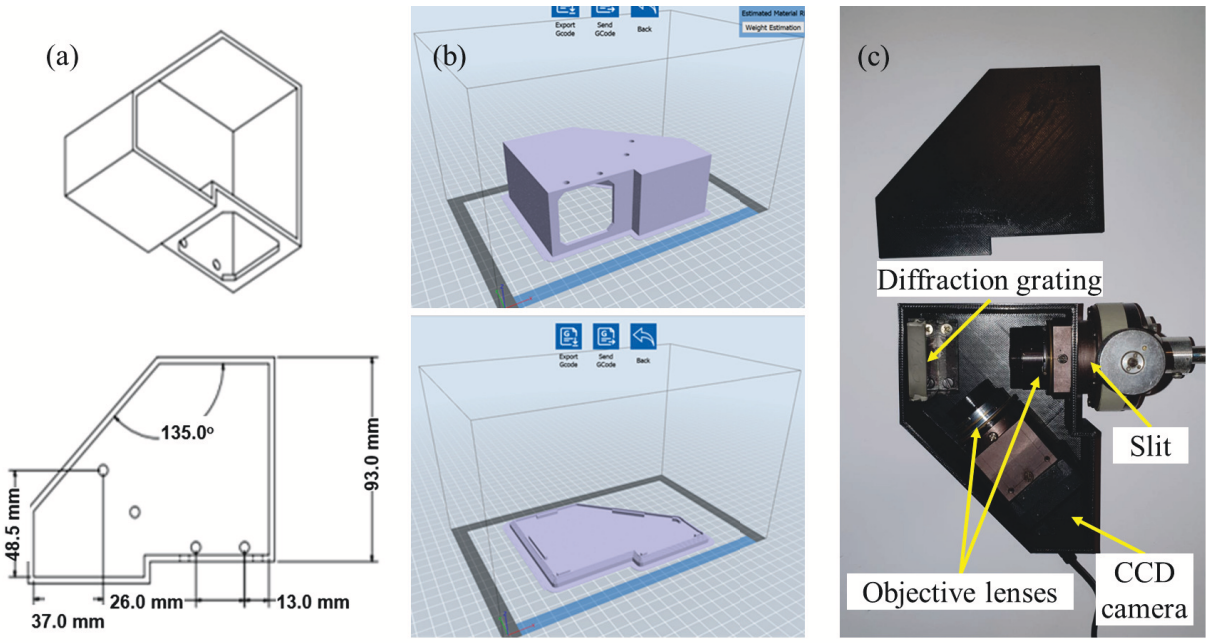

Figure 3. Mechanical design of the optical spectrometer structure (a), SolidWorks design (b) and the mounted optical spectrometer (c).

3. Electronic power supply and processing software

The electronic power supply and control circuit of the pulsed laser was fully designed and built using commercially available electronics, thus reducing the costs. In addition, we developed a customized software for the acquisition and processing of the optical signals produced by the plasma.

3.1. Electronic power supply

In previous work [20], we developed an electronic system for power supply and control of the laser head, which had a limited rate of successful electronic signal triggering a laser pulse (of around 60%). Furthermore, the laser worked in a passive Q-switch regime that generated several laser pulses for

each successful trigger (i.e., the total laser energy was distributed between those pulses, thus reducing the effectiveness of the LIBS). A Cr^{4+}:YAG saturable crystal was used for the passive Q-switch regime. In this paper, we optimized the power supply electronics by gradually increasing the energy applied to the laser until the optimal threshold was reached (545 VDC, energy input of 16 J) for the optical pumping. This allows to obtain a single laser pulse by controlling the energy parameters that are sent to the laser head, while keeping the passive Q-switch regime.
The maximum input energy used for pumping the flash lamp was 28 J at 720 VDC (see the oscillogram in Figure 4(a)), resulting in 4 pulses at 150 µs flash lamp pulse duration. By decreasing the input energy of the flash lamp, we obtained a reduction in the number of pulses observed in this regime. Figure 4(b) shows the oscillogram of the obtained monopulse with an energy input of 16 J.

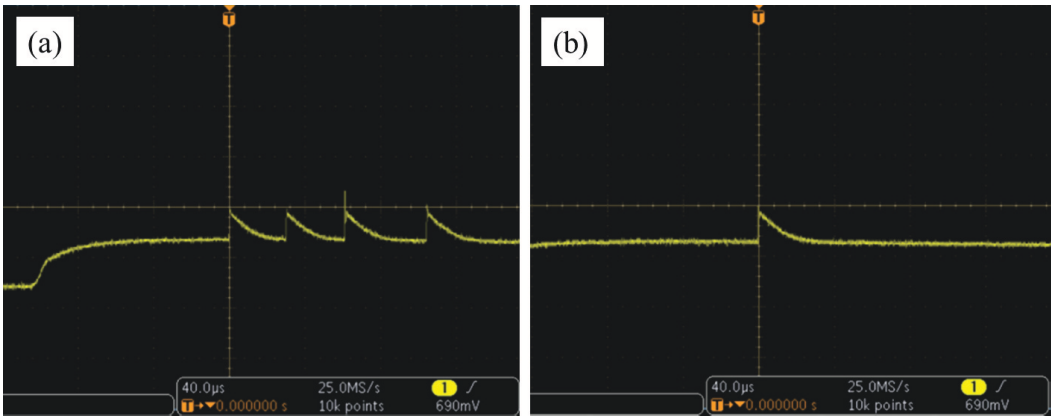

Figure 4. Oscillogram of the laser pulses in the passive Q-switch regime: four pulses are obtained with an energy input of 28 J (a), and a monopulse is obtained with an energy input of 16 J (b).

The new power supply electronics has three main components: (a) the power circuit, (b) the triggering circuit, and (c) the control circuit. The electronic circuits (a) and (b) were designed in the Proteus[1] (see Figure 5), and the corresponding PCBs (printed circuit boards) were manufactured on a CNC (computer numerical control). Figure 6 shows the assembled power circuit. In contrast, the control circuit (c) uses an Arduino Uno microcontroller which controls the stages of operation of the power circuit (a) and the triggering circuit (b).
Additionally, in order to ensure a higher success rate of the laser pulse triggering (from around 60% to 99%), we apply the following optimizations in the electronic design:

- We avoid wired electrical connections between the three main aforementioned circuits of the power supply electronics, and use optical connections instead (optocouplers), in order to eliminate any electrical noise and high-voltage interference between the circuits.
- We use capacitors to store the electrical energy in the power circuit, required to activate the laser head. These capacitors are charged only when the laser is triggered, instead of a continuous charge of our previous design [20].

[1] https://www.labcenter.com/pcb/

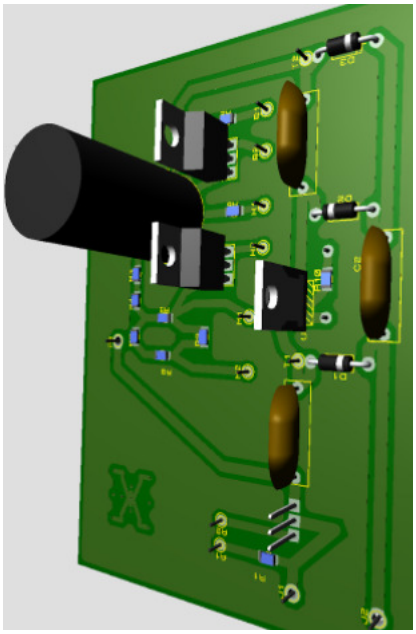

Figure 5. Design of the electronic circuit in the Proteus PCB Design software.

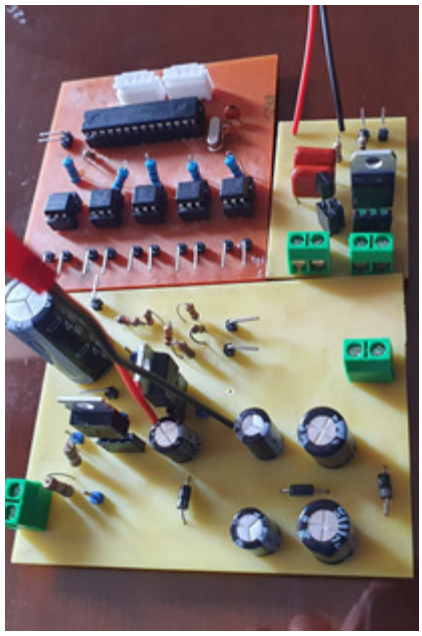

Figure 6. The assembled power circuit

All the circuits of the LIBS electronic system were assembled and inserted into a compact 3D-printed case.

3.2. Processing software

We developed a custom software for the acquisition and processing of the spectrum produced by the analyzed samples. The software also controls the Chameleon CCD digital camera and constantly captures the light images (every 55 ms) after the laser radiation generates plasma in the sample. The software detects any drastic change in the intensity of the captured light, and stores the images. Then, the software processes the captured image and generates the corresponding spectral lines. Note that the Chameleon CCD digital camera used in the device, has a minimum trigger delay time of 0.1 ms, which is too high compared to the ideal delay of 500 ns, to capture the image after the generation of plasma [2]. This ideal delay is critical for quantification of atomic elements, however, since our current work focuses on identification of atomic elements, the proposed solution based on constant image capture is adapted to this purpose. Adding a fast triggering electronic control system to synchronize the laser pulse with the image capture may increase the cost of the device. Such a development will be covered in future work.

The developed software has an embedded database with reference spectral information (peak values of the main spectral lines) of several elements of interest from the periodic table. These references were obtained from the National Institute of Standards and Technology (NIST)[2]. Our software compares this database with the spectra obtained by the proposed LIBS system, by applying a peak detection algorithm. This allows qualitatively identifying the existence of the element of interest in the analyzed sample and a visual comparison between the generated spectra and the references.

Figure 7 shows the main window of the control panel of our LIBS software. The software allows users to observe the images captured from the camera in real-time. It also allows the calibration of the spectrometer as a function of the wavelength. Finally, it is possible to select elements of the database

[2] https://www.nist.gov/pml/atomic-spectra-database/

and perform the processing of the images to obtain the spectral lines of the analyzed samples and compare them with the referential data.

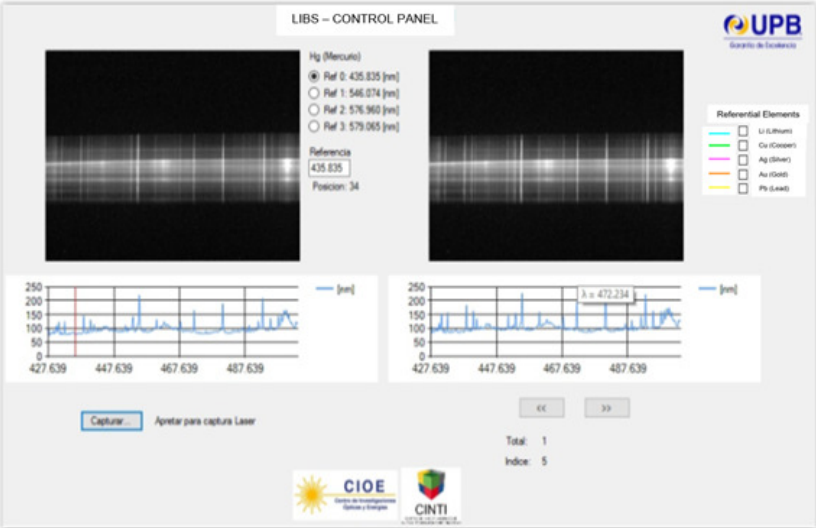

Figure 7. The LIBS control software.

4. Results

The proposed compact and low-cost 3D-printed portable LIBS system (shown in Figure 8) includes an optical spectrometer, an electronic power supply with a control of the Nd^{3+}:YAG pulsed laser, and a handheld measuring device that collects the light produced by the plasma on the sample and sends it through an optical fiber to the spectrometer. The dimensions of the different components of the device are: 14.5 cm X 15.4 cm X 12 cm for the handheld measuring system, 16.8 cm X 11.8 cm X 11.5 cm for the power supply case, and 13.2 cm X 10.4 X 9.60 cm for the spectrometer.

Thanks to the improvements made in electronics, mechanical design, robustness and portability, our LIBS system becomes a Technology Readiness Level 5 (TRL 5) equipment [14]. The TRL 5 category confers the status of functional equipment for a real environment. This feature is very important to be able to use our system in different application areas of the LIBS technique that require non-destructive, real-time and in-situ analysis.

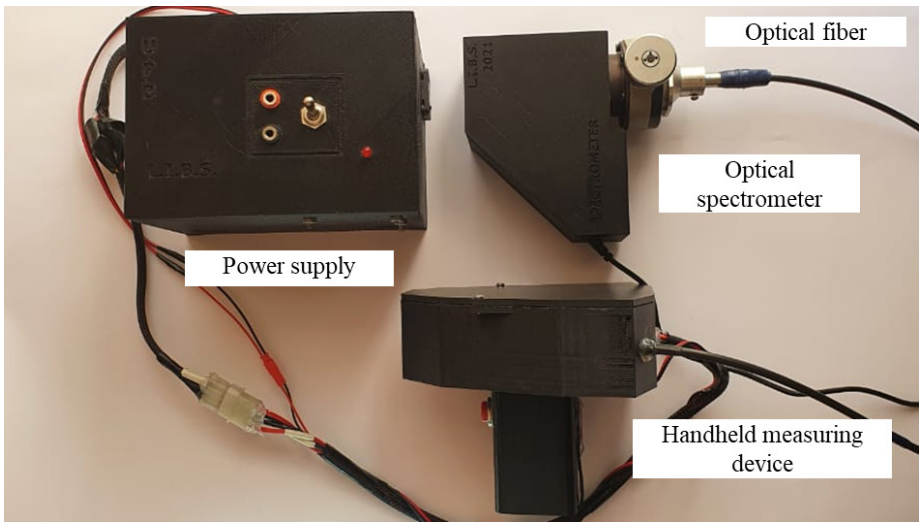

Figure 8. The complete proposed 3D-printed portable LIBS system.

To validate the proposed LIBS system, we carried out several tests with samples of atomic elements (i.e., copper, lithium, silver, gold and lead). Figure 9 shows spectral lines of copper and lithium where laser shots were made with the handheld measuring device in the real samples.
In the control software, we select the atomic element to be analyzed (i.e., Ag, Li) from the database and the corresponding reference lines are displayed over the obtained spectrum. The software compares the peaks of the obtained spectrum against the reference lines from the database (i.e. wavelengths λ_1=521.06 nm and λ_2=546.70 nm for silver; and λ_1=610.50 nm for Li). Figures 9(a) and 9(b) clearly show the coincidence of the spectral lines obtained with our LIBS system with those of the reference elements.

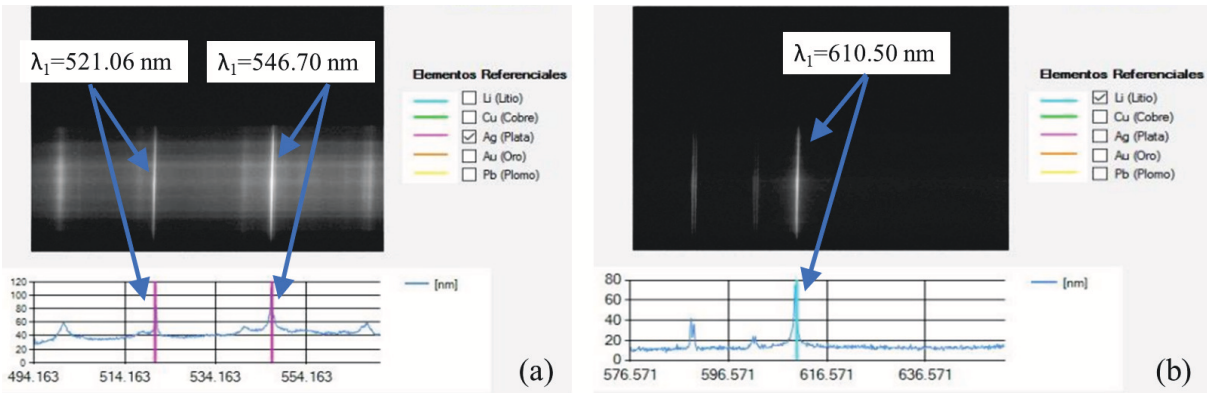

Figure 9. Spectral lines of real samples of silver (a) and the spectral line of lithium (b).

The experimental results with real samples demonstrate the validity of the proposed LIBS system for the identification of atomic elements of interest in real-time.

5. Conclusions

In this paper we present the development of a compact low-cost 3D-printed portable LIBS system, for the identification of atomic elements in-situ and in real-time. This development allowed us to improve our LIBS technology, evolving from a TRL 4 (laboratory validation) to a TRL 5 (validation in relevant environment) category. This step was possible thanks to: (a) the optimization of the control electronic system to avoid pulse losses during laser triggering, thanks to the use of optical connections (optocouplers) avoiding wired electrical connections of the circuits to eliminate any electrical noise, (b) efficient single pulse generation of the passive Q-switch regime though an energy control of the optical pumping, and (c) a robust and portable 3D-printed system, validated with real samples of different elements in a relevant environment. The estimated cost of the proposed LIBS device is around 2000 USD.
With this successful experience, our plan for future work is to reach TRL 6, i.e., demonstrate our technology in relevant environments, such as the mining industry or archeology. Also in this stage, an electronic trigger control system will be implemented to synchronize the laser pulse and the capture of the optical signals.

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